Lending Club Loans — Data Description

Accompanies loans-kpi.csv | 45-952 End-to-End Business Analytics | Fall 2026

This file describes the fields in **loans-kpi.csv**, the cleaned data set supplied for the “Screening peer-lending loans for better outcomes” case (Lending Club). Each row represents one 3-year loan funded through the LendingClub.com platform between May 2007 and February 2010. The data is adapted from the online resources for *The Analytics Edge* by Bertsimas, O'Hair, and Pulleyblank (Dynamic Ideas LLC, 2016).

Data set summary

Rows (loans)	9,578
Columns	17 (13 borrower/loan fields + 1 target + 3 computed KPI fields)
Unit of analysis	One 3-year loan
Target variable	default (0 / 1)
Class balance	16.0% default / 84.0% paid in full
Missing values	None.

Field dictionary

Borrower and loan attributes (available to the investor when deciding whether to fund a loan)

Field	Type / values	Description
credit.policy	0 / 1	1 if the customer meets the credit underwriting criteria of LendingClub.com, and 0 otherwise. 80.5% of loans meet the criteria.
purpose	credit_card / debt_consolidation / educational / major_purchase / small_business / home_improvement / all_other	The stated purpose of the loan.
int.rate	Proportion, 0.0600–0.2164	The interest rate of the loan (e.g., a rate of 11% is stored as 0.11). Borrowers judged by LendingClub.com to be more risky are assigned higher interest rates. Mean 12.3%.
installment	USD, 15.67–940.14	The monthly installment owed by the borrower if the loan is funded. Mean $319.09.
log.annual.inc	Numeric, 7.5475–14.5284	The natural log of the self-reported annual income of the borrower.
dti	Numeric, 0.00–29.96	The debt-to-income ratio of the borrower (amount of debt divided by annual income).
fico	Integer, 612–827	The FICO credit score of the borrower. Mean 711.
days.with.cr.line	Numeric, 178.96–17,639.96	The number of days the borrower has had a credit line.
revol.bal	USD, 0–1,207,359	The borrower's revolving balance (amount unpaid at the end of the credit card billing cycle). Heavily right-skewed — worth screening for extreme values.

Field	Type / values	Description
revol.util	Percent, 0.0–119.0	The borrower's revolving line utilization rate (the amount of the credit line used relative to total credit available). Values can exceed 100 for borrowers over their nominal limit.
inq.last.6mths	Integer, 0–33	The borrower's number of inquiries by creditors in the last 6 months.
delinq.2yrs	Integer, 0–13	The number of times the borrower had been 30+ days past due on a payment in the past 2 years.
pub.rec	Integer, 0–5	The borrower's number of derogatory public records (bankruptcy filings, tax liens, or judgments).

Target

Field	Type / values	Description
default	0 / 1	1 if the loan was not paid back in full — the borrower either defaulted or the loan was “charged off” (deemed unlikely to ever be paid back). This is the outcome to be predicted (16.0% = 1).

Computed KPI fields (added after running the first-week script)

Field	Type / values	Description
tot.payment	USD, 564.12–33,845.04	The total payments from the loan, computed as 36 monthly installments (installment × 36).
principal	USD, 410.83–21,844.39	The principal value, computed by dividing total payments by (1 + compounded interest rate), well approximated by tot.payment / exp(3 × int.rate).
interest	USD, 120.37–15,743.58	The interest payments that accrue to the investor (tot.payment – principal).

Notes for analysis

• **Profit and loss:** for non-default loans, the investor's profit is the interest earned; for default loans, the investor's loss is the principal value. These can be used to compute net profits and losses and other derived metrics like return on invested capital (ROIC), which are central to the Update 3 optimization exercise.

• **Correlated fields:** int.rate is set by LendingClub as a function of assessed risk, so it is likely correlated with fico, revol.util, and other risk indicators — examine these correlations before finalizing your logistic regression variable set (Update 2).

• **Data dynamism:** credit.policy, purpose, int.rate, installment, and the borrower credit-bureau fields (log.annual.inc, dti, fico, days.with.cr.line, revol.bal, revol.util, inq.last.6mths, delinq.2yrs, pub.rec) are all known at the time of the funding decision. tot.payment, principal, and interest are computed only after the loan's outcome is known and should not be used as predictive features — they are provided for calculating profit, loss, and ROIC once a screening decision has been made.

• **Source and license:** adapted from the online resources for *The Analytics Edge* by Bertsimas, O'Hair, and Pulleyblank (Dynamic Ideas LLC, 2016), based on data that used to be publicly available from Lending Club. Provided for classroom use in 45-952 only.